

NestedFlow Automation Setup Instructions

Creation Date: 06/08/2025

Table Of Contents

Introduction	1
Tool Setup	2
.NET Runtime Setup	6
Web Driver Setup	7
Pre Requisites:	7
Installing Chrome Driver.....	8
Installing Edge Driver	9
Java legacy applications Setup (Optional)	10
Appium Setups (Optional)	14
GitHub Setup (Optional)	15
Code Considerations (optional)	22

Introduction

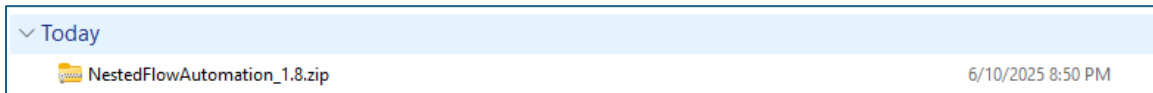
This document explains the initial setups needed to make NestedFlow Automation tool fully functional for all its intended capabilities. This involves:

- Tool Setup
- .NET Runtime Setup
- Web Driver Setup
- Java legacy applications (optional)
- Appium Setups (optional)
- GitHub Setup (Optional)
- Code Considerations (optional)

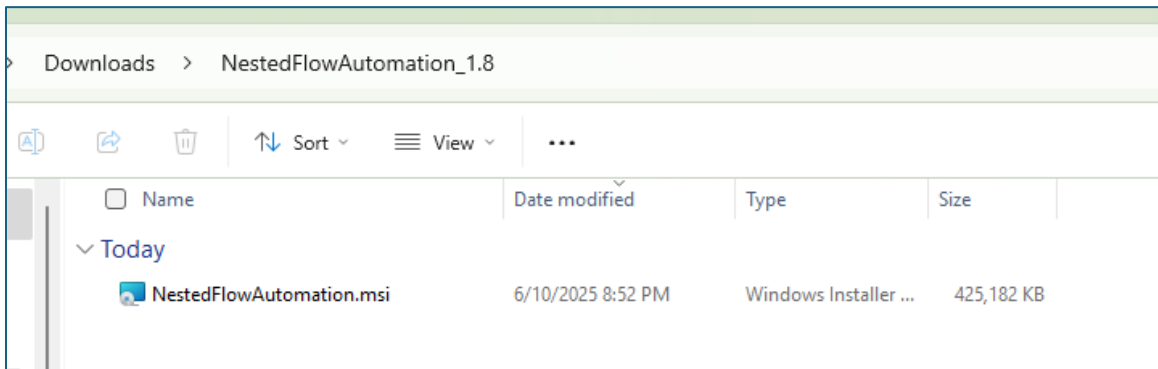
Tool Setup

1. Download the latest version from the location:

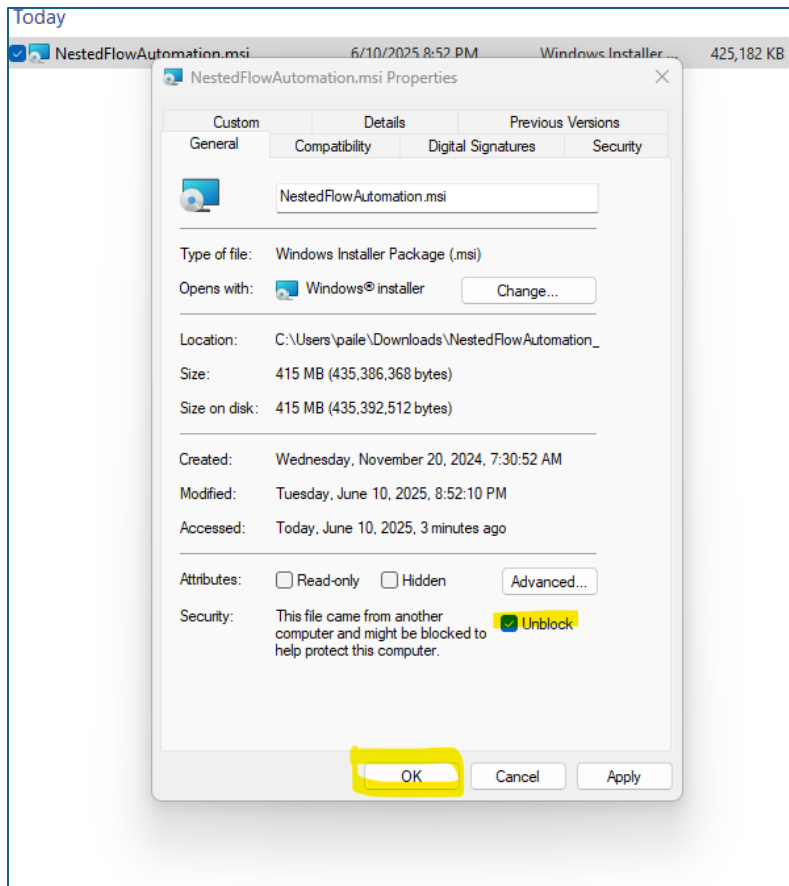
https://sourceforge.net/projects/nestedflowautomation/files/NestedFlowAutomation_1.8.zip/download



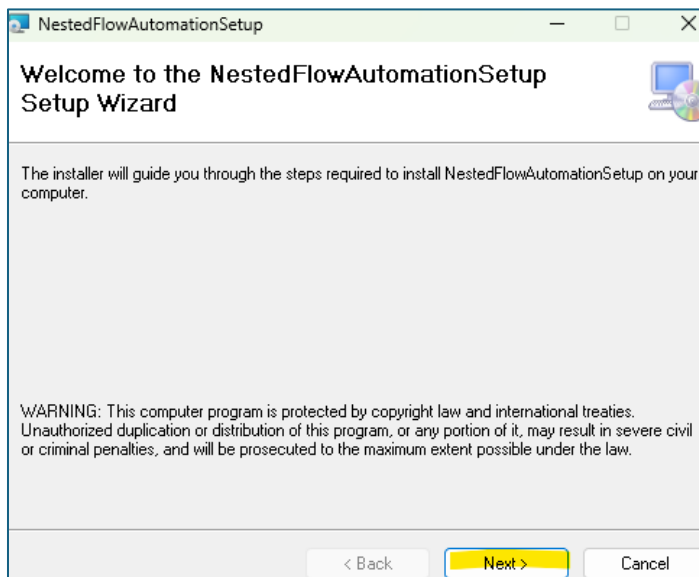
2. Unzip the file

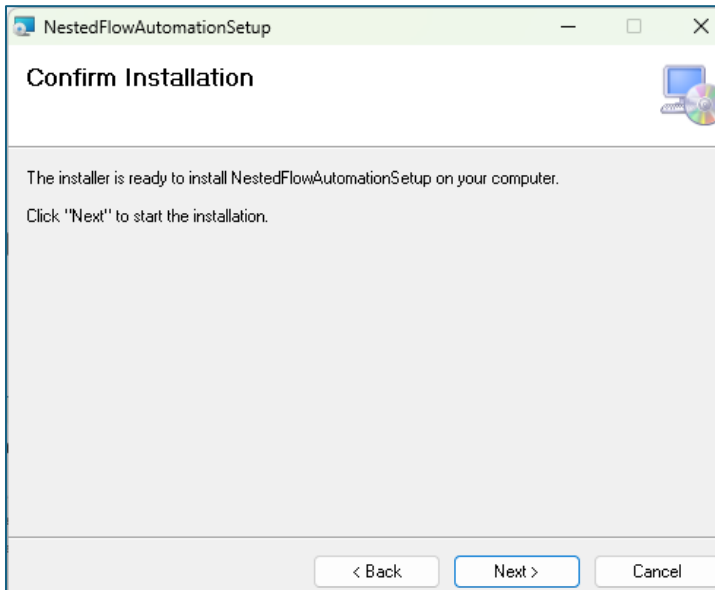
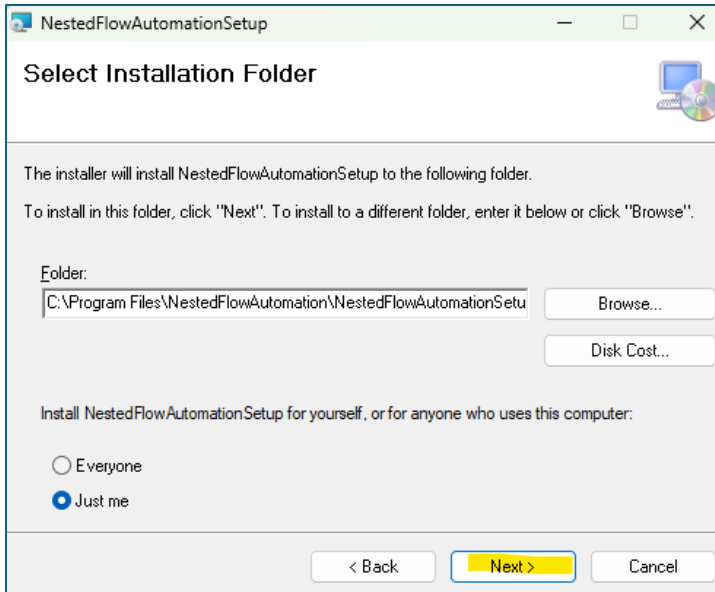


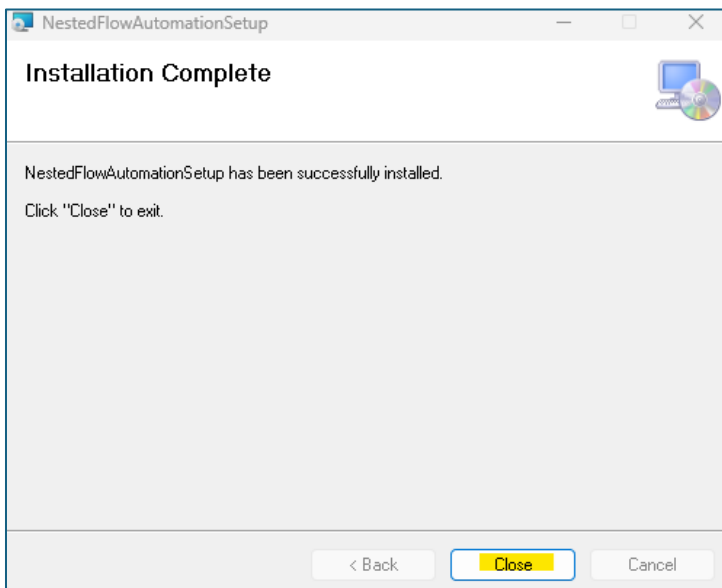
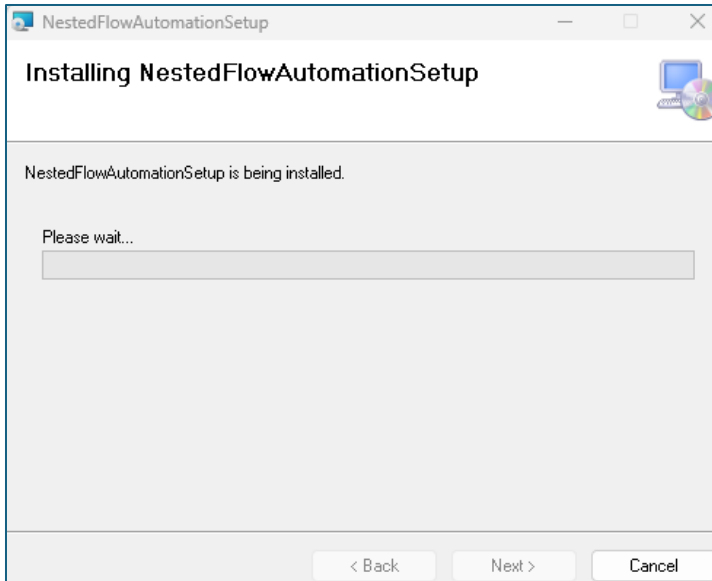
3. Right click on the file and choose properties and **Unblock** the file



4. Double click on the MSI to complete the installation







.NET Runtime Setup

NestedFlow Automation version 1.7 is built with .NET 6 and is a 32-bit application. To run the application we need to download and install .NET6 runtime: [Download .NET 6.0 \(Linux, macOS, and Windows\)](#) | [.NET](#)

Download .NET 6.0

 This release has reached end of life, meaning it is no longer supported. We recommend moving to a supported release, such as [.NET 9.0](#). See our [support policy](#) for more details.

^ 6.0.36

[Release notes](#) Latest release date November 12, 2024

Build apps - SDK 

SDK 6.0.428

OS	Installers	Binaries
Linux	Package manager instructions	Arm32 Arm32 Alpine Arm64 Arm64 Alpine x64 x64 Alpine
macOS	Arm64 x64	Arm64 x64
Windows	x64 x86 Arm64 winget instructions	x64 x86 Arm64
All	dotnet-install scripts	

Visual Studio support
Visual Studio 2022 (v17.3)

Included runtimes
.NET Runtime 6.0.36
ASP.NET Core Runtime 6.0.36
.NET Desktop Runtime 6.0.36

Language support
C# 10.0
F# 6.0

Run apps - Runtime 

ASP.NET Core Runtime 6.0.36

The ASP.NET Core Runtime enables you to run existing web/server applications. **On Windows, we recommend installing the Hosting Bundle, which includes the .NET Runtime and IIS support.**

IIS runtime support (ASP.NET Core Module v2)
16.0.24290.36

OS	Installers	Binaries
Linux	Package manager instructions	Arm32 Arm32 Alpine Arm64 Arm64 Alpine x64 x64 Alpine
macOS		Arm64 x64
Windows	x64 x86 Hosting Bundle winget instructions	x64 x86 Arm64

.NET Desktop Runtime 6.0.36

The .NET Desktop Runtime enables you to run existing Windows desktop applications. **This release includes the .NET Runtime; you don't need to install it separately.**

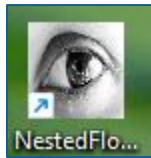
Web Driver Setup

NestedFlow Automation tool has the capability to auto download the Chrome and Edge browser drivers.

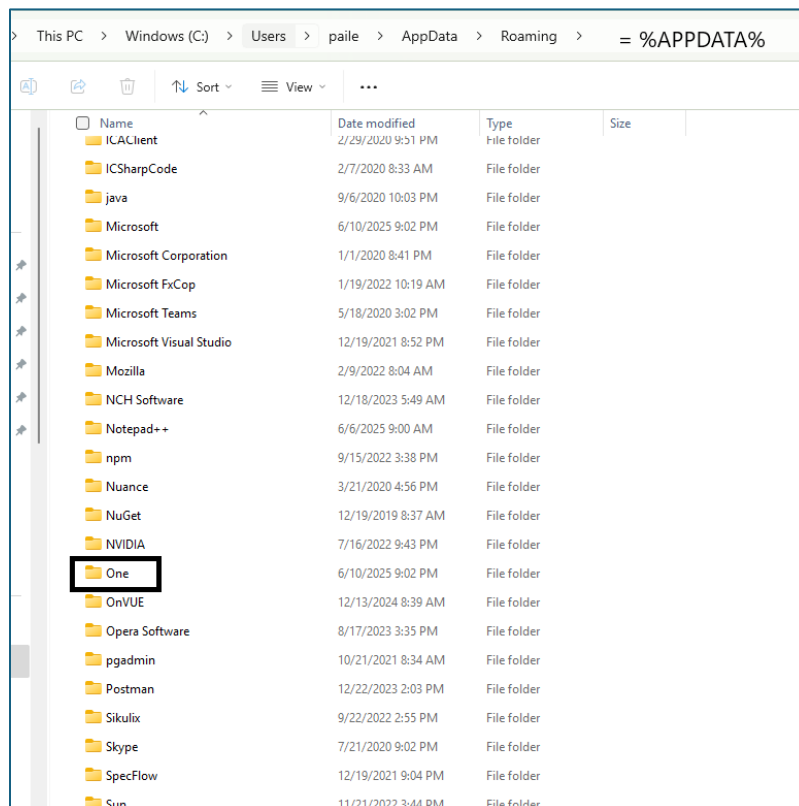
Pre Requisites:

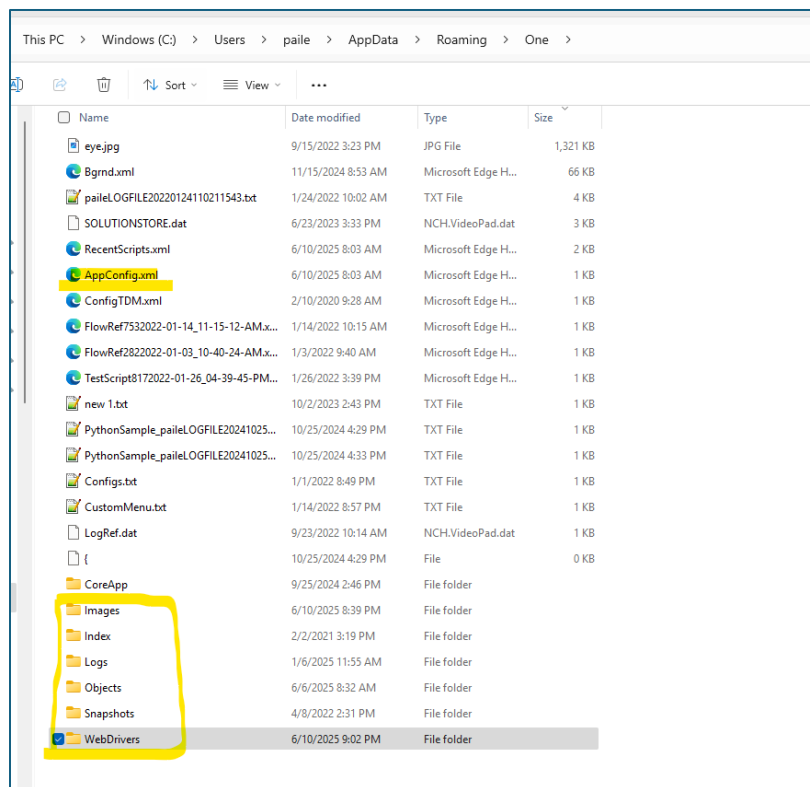
- Chrome and Edge drivers should be installed on the browser
- Firewall connectivity between the machine and below end points should be open:
 - <https://msedgedriver.azureedge.net>
 - <https://storage.googleapis.com>

Once the setup is successfully completed, a desktop icon will be created:

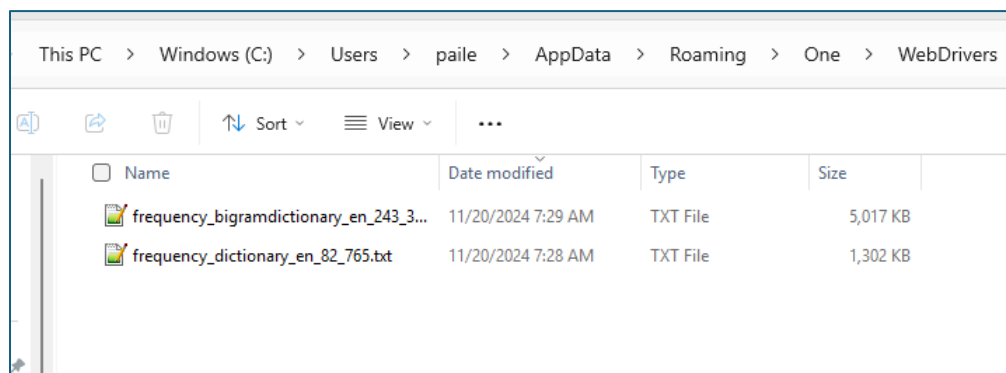


Double click on this icon to open the tool. When the tool is opened for the first time, below folder structure is created in %APPDATA% folder



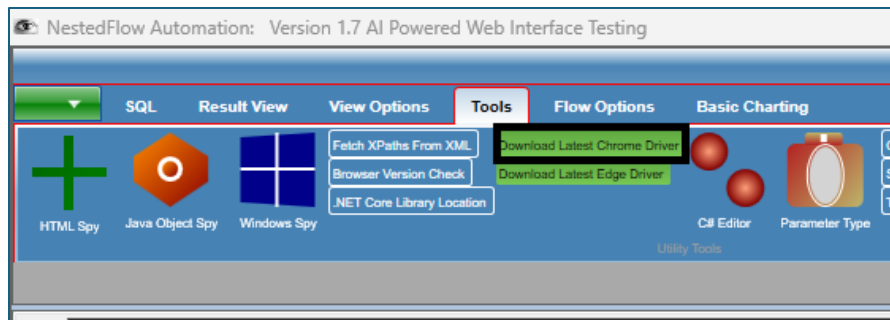


Open the **WebDrivers** folder

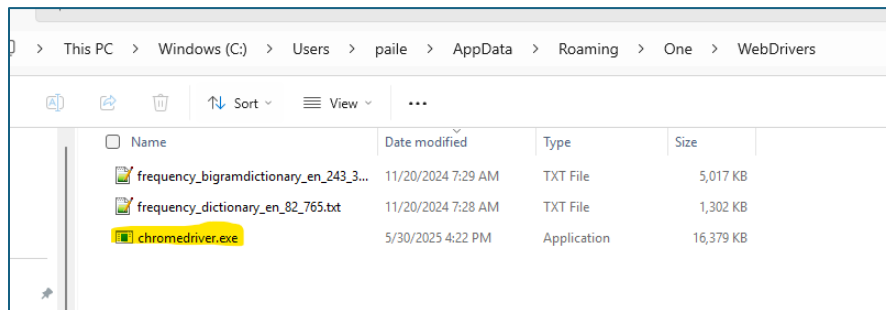


Installing Chrome Driver

On the **Tools** menu click on **Download Latest Chrome Driver**



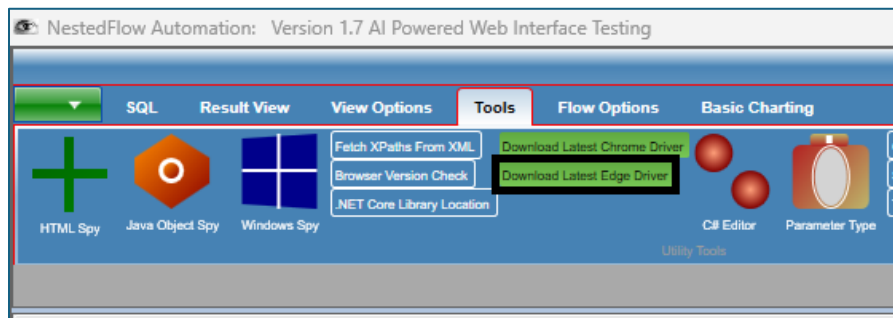
Chrome Driver associated with the installed browser will get downloaded



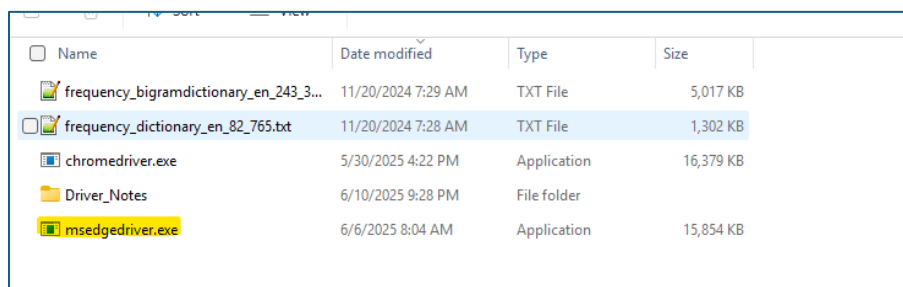
Note: Please perform this step every time the Chrome browser upgrades

Installing Edge Driver

On the **Tools** menu click on **Download Latest Edge Driver**



Edge Driver associated with the installed browser will get downloaded

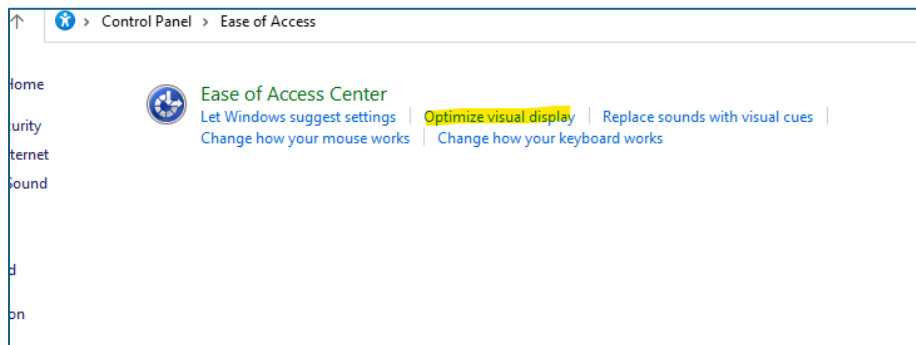
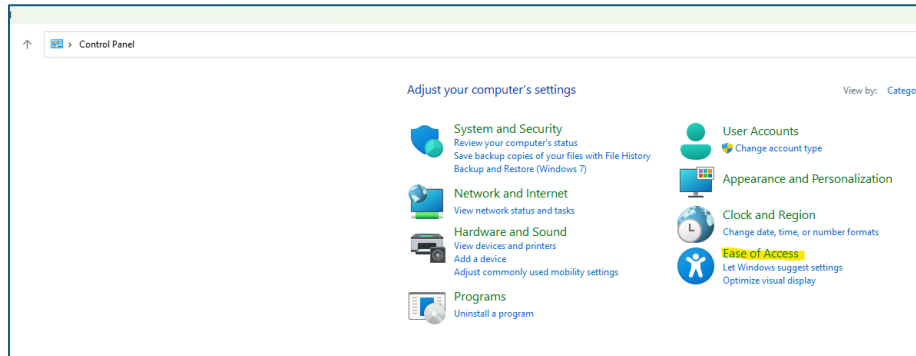


Note: Please perform this step every time the Edge browser upgrades

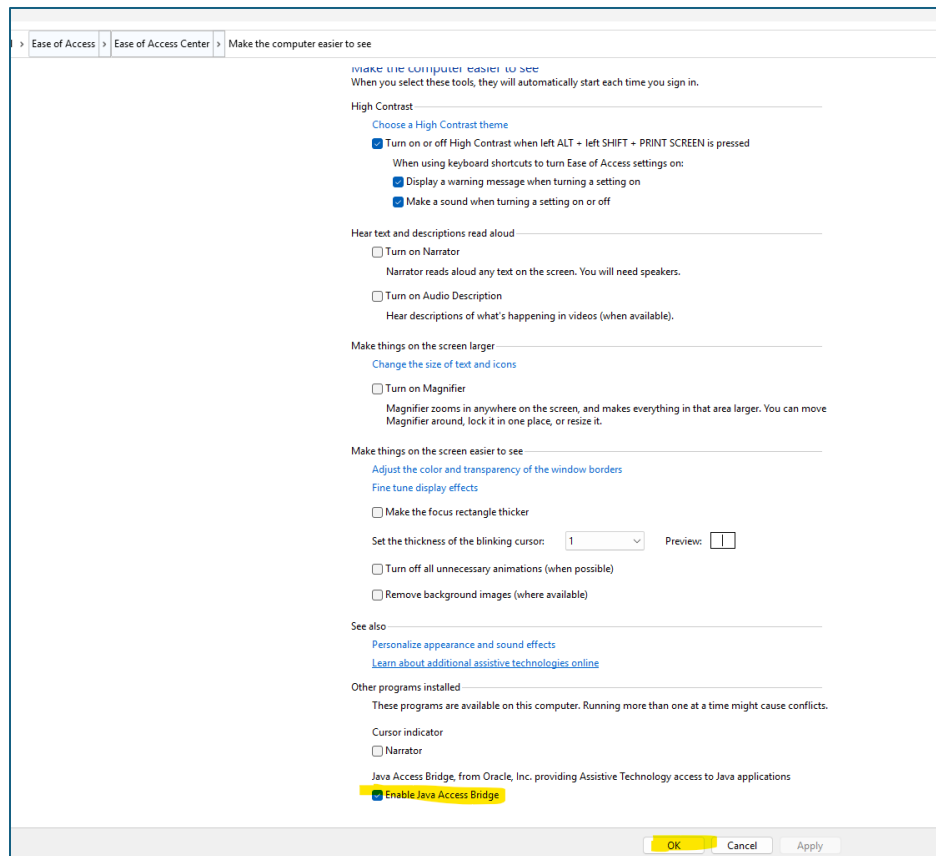
Java legacy applications Setup (Optional)

Pre Requisite: Java should be setup on the machine

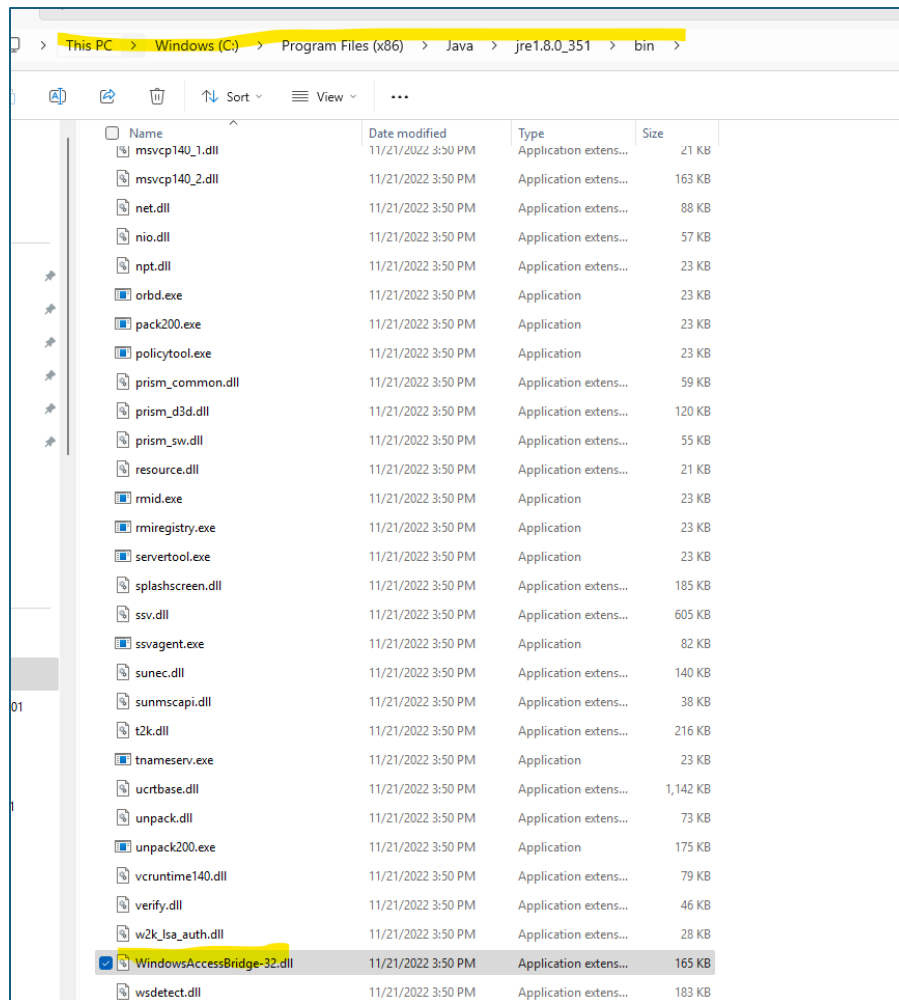
In the Control Panel click on **Ease Of Access**



Click on Optimize Visual Display



Click on **Enable Java Access Bridge** and click on **Ok** button



Copy the WindowsAccessBridge-32.dll from %JAVA_HOME%/bin folder to tool exe folder in **Program files**

× + 🏠 This PC Windows (C:) Program Files NestedFlowAutomation NestedFlowAutomationSetup																																																																																																																																																																				
📁 🔍 🗑️ ⬆️⬆️ Sort ▾ 📄 View ▾ ⋮																																																																																																																																																																				
<table> <tr> <th>📁 Name</th><th>Date modified</th><th>Type</th><th>Size</th><th></th></tr> <tr> <td>System.Web.Services.Description.dll</td><td>11/20/2024 1:30 AM</td><td>Application extens...</td><td>3/3 KB</td><td></td></tr> <tr> <td>System.Windows.Extensions.dll</td><td>11/20/2024 7:29 AM</td><td>Application extens...</td><td>26 KB</td><td></td></tr> <tr> <td>System.Windows.Forms.DataVisualization...</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>1,643 KB</td><td></td></tr> <tr> <td>TDAPIOLELib.dll</td><td>11/20/2024 7:27 AM</td><td>Application extens...</td><td>1,452 KB</td><td></td></tr> <tr> <td>Tesseract.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>124 KB</td><td></td></tr> <tr> <td>UglyToad.PdfPig.Core.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>45 KB</td><td></td></tr> <tr> <td>UglyToad.PdfPig.dll</td><td>11/20/2024 7:27 AM</td><td>Application extens...</td><td>4,097 KB</td><td></td></tr> <tr> <td>UglyToad.PdfPig.DocumentLayoutAn...</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>243 KB</td><td></td></tr> <tr> <td>UglyToad.PdfPig.Fonts.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>1,058 KB</td><td></td></tr> <tr> <td>UglyToad.PdfPig.Package.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>5 KB</td><td></td></tr> <tr> <td>UglyToad.PdfPig.Tokenization.dll</td><td>11/20/2024 7:27 AM</td><td>Application extens...</td><td>24 KB</td><td></td></tr> <tr> <td>UglyToad.PdfPig.Tokens.dll</td><td>11/20/2024 7:27 AM</td><td>Application extens...</td><td>43 KB</td><td></td></tr> <tr> <td>Web Automation Framework.deps.json</td><td>11/20/2024 7:29 AM</td><td>JSON File</td><td>1 KB</td><td></td></tr> <tr> <td>Web Automation Framework.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>459 KB</td><td></td></tr> <tr> <td>Web Automation Framework.pdb</td><td>11/20/2024 7:29 AM</td><td>VisualStudio.pdb....</td><td>171 KB</td><td></td></tr> <tr> <td>WebDriver.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>5,861 KB</td><td></td></tr> <tr> <td>WebDriver.Support.dll</td><td>11/20/2024 7:29 AM</td><td>Application extens...</td><td>35 KB</td><td></td></tr> <tr> <td>WindowsAutomation.deps.json</td><td>11/20/2024 7:28 AM</td><td>JSON File</td><td>10 KB</td><td></td></tr> <tr> <td>WindowsAutomation.dll</td><td>11/20/2024 7:27 AM</td><td>Application extens...</td><td>95 KB</td><td></td></tr> <tr> <td>WindowsAutomation.pdb</td><td>11/20/2024 7:28 AM</td><td>VisualStudio.pdb....</td><td>59 KB</td><td></td></tr> <tr> <td>WPFTxtEditor.deps.json</td><td>11/20/2024 7:30 AM</td><td>JSON File</td><td>1 KB</td><td></td></tr> <tr> <td>WPFTxtEditor.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>38 KB</td><td></td></tr> <tr> <td>WPFTxtEditor.pdb</td><td>11/20/2024 7:28 AM</td><td>VisualStudio.pdb....</td><td>22 KB</td><td></td></tr> <tr> <td>WPFXMLParser.deps.json</td><td>11/20/2024 7:28 AM</td><td>JSON File</td><td>1 KB</td><td></td></tr> <tr> <td>WPFXMLParser.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>1,067 KB</td><td></td></tr> <tr> <td>WPFXMLParser.pdb</td><td>11/20/2024 7:29 AM</td><td>VisualStudio.pdb....</td><td>162 KB</td><td></td></tr> <tr> <td>zing.dll</td><td>11/20/2024 7:27 AM</td><td>Application extens...</td><td>472 KB</td><td></td></tr> <tr> <td>Zxing.Net.Core.dll</td><td>11/20/2024 7:27 AM</td><td>Application extens...</td><td>70 KB</td><td></td></tr> <tr> <td>Zxing.Windows.Compatibility.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>33 KB</td><td></td></tr> <tr> <td>zing-core.dll</td><td>11/20/2024 7:28 AM</td><td>Application extens...</td><td>376 KB</td><td></td></tr> <tr> <td>WindowsAccessBridge-32.dll</td><td>11/21/2022 3:50 PM</td><td>Application extens...</td><td>165 KB</td><td></td></tr> </table>					📁 Name	Date modified	Type	Size		System.Web.Services.Description.dll	11/20/2024 1:30 AM	Application extens...	3/3 KB		System.Windows.Extensions.dll	11/20/2024 7:29 AM	Application extens...	26 KB		System.Windows.Forms.DataVisualization...	11/20/2024 7:28 AM	Application extens...	1,643 KB		TDAPIOLELib.dll	11/20/2024 7:27 AM	Application extens...	1,452 KB		Tesseract.dll	11/20/2024 7:28 AM	Application extens...	124 KB		UglyToad.PdfPig.Core.dll	11/20/2024 7:28 AM	Application extens...	45 KB		UglyToad.PdfPig.dll	11/20/2024 7:27 AM	Application extens...	4,097 KB		UglyToad.PdfPig.DocumentLayoutAn...	11/20/2024 7:28 AM	Application extens...	243 KB		UglyToad.PdfPig.Fonts.dll	11/20/2024 7:28 AM	Application extens...	1,058 KB		UglyToad.PdfPig.Package.dll	11/20/2024 7:28 AM	Application extens...	5 KB		UglyToad.PdfPig.Tokenization.dll	11/20/2024 7:27 AM	Application extens...	24 KB		UglyToad.PdfPig.Tokens.dll	11/20/2024 7:27 AM	Application extens...	43 KB		Web Automation Framework.deps.json	11/20/2024 7:29 AM	JSON File	1 KB		Web Automation Framework.dll	11/20/2024 7:28 AM	Application extens...	459 KB		Web Automation Framework.pdb	11/20/2024 7:29 AM	VisualStudio.pdb....	171 KB		WebDriver.dll	11/20/2024 7:28 AM	Application extens...	5,861 KB		WebDriver.Support.dll	11/20/2024 7:29 AM	Application extens...	35 KB		WindowsAutomation.deps.json	11/20/2024 7:28 AM	JSON File	10 KB		WindowsAutomation.dll	11/20/2024 7:27 AM	Application extens...	95 KB		WindowsAutomation.pdb	11/20/2024 7:28 AM	VisualStudio.pdb....	59 KB		WPFTxtEditor.deps.json	11/20/2024 7:30 AM	JSON File	1 KB		WPFTxtEditor.dll	11/20/2024 7:28 AM	Application extens...	38 KB		WPFTxtEditor.pdb	11/20/2024 7:28 AM	VisualStudio.pdb....	22 KB		WPFXMLParser.deps.json	11/20/2024 7:28 AM	JSON File	1 KB		WPFXMLParser.dll	11/20/2024 7:28 AM	Application extens...	1,067 KB		WPFXMLParser.pdb	11/20/2024 7:29 AM	VisualStudio.pdb....	162 KB		zing.dll	11/20/2024 7:27 AM	Application extens...	472 KB		Zxing.Net.Core.dll	11/20/2024 7:27 AM	Application extens...	70 KB		Zxing.Windows.Compatibility.dll	11/20/2024 7:28 AM	Application extens...	33 KB		zing-core.dll	11/20/2024 7:28 AM	Application extens...	376 KB		WindowsAccessBridge-32.dll	11/21/2022 3:50 PM	Application extens...	165 KB	
📁 Name	Date modified	Type	Size																																																																																																																																																																	
System.Web.Services.Description.dll	11/20/2024 1:30 AM	Application extens...	3/3 KB																																																																																																																																																																	
System.Windows.Extensions.dll	11/20/2024 7:29 AM	Application extens...	26 KB																																																																																																																																																																	
System.Windows.Forms.DataVisualization...	11/20/2024 7:28 AM	Application extens...	1,643 KB																																																																																																																																																																	
TDAPIOLELib.dll	11/20/2024 7:27 AM	Application extens...	1,452 KB																																																																																																																																																																	
Tesseract.dll	11/20/2024 7:28 AM	Application extens...	124 KB																																																																																																																																																																	
UglyToad.PdfPig.Core.dll	11/20/2024 7:28 AM	Application extens...	45 KB																																																																																																																																																																	
UglyToad.PdfPig.dll	11/20/2024 7:27 AM	Application extens...	4,097 KB																																																																																																																																																																	
UglyToad.PdfPig.DocumentLayoutAn...	11/20/2024 7:28 AM	Application extens...	243 KB																																																																																																																																																																	
UglyToad.PdfPig.Fonts.dll	11/20/2024 7:28 AM	Application extens...	1,058 KB																																																																																																																																																																	
UglyToad.PdfPig.Package.dll	11/20/2024 7:28 AM	Application extens...	5 KB																																																																																																																																																																	
UglyToad.PdfPig.Tokenization.dll	11/20/2024 7:27 AM	Application extens...	24 KB																																																																																																																																																																	
UglyToad.PdfPig.Tokens.dll	11/20/2024 7:27 AM	Application extens...	43 KB																																																																																																																																																																	
Web Automation Framework.deps.json	11/20/2024 7:29 AM	JSON File	1 KB																																																																																																																																																																	
Web Automation Framework.dll	11/20/2024 7:28 AM	Application extens...	459 KB																																																																																																																																																																	
Web Automation Framework.pdb	11/20/2024 7:29 AM	VisualStudio.pdb....	171 KB																																																																																																																																																																	
WebDriver.dll	11/20/2024 7:28 AM	Application extens...	5,861 KB																																																																																																																																																																	
WebDriver.Support.dll	11/20/2024 7:29 AM	Application extens...	35 KB																																																																																																																																																																	
WindowsAutomation.deps.json	11/20/2024 7:28 AM	JSON File	10 KB																																																																																																																																																																	
WindowsAutomation.dll	11/20/2024 7:27 AM	Application extens...	95 KB																																																																																																																																																																	
WindowsAutomation.pdb	11/20/2024 7:28 AM	VisualStudio.pdb....	59 KB																																																																																																																																																																	
WPFTxtEditor.deps.json	11/20/2024 7:30 AM	JSON File	1 KB																																																																																																																																																																	
WPFTxtEditor.dll	11/20/2024 7:28 AM	Application extens...	38 KB																																																																																																																																																																	
WPFTxtEditor.pdb	11/20/2024 7:28 AM	VisualStudio.pdb....	22 KB																																																																																																																																																																	
WPFXMLParser.deps.json	11/20/2024 7:28 AM	JSON File	1 KB																																																																																																																																																																	
WPFXMLParser.dll	11/20/2024 7:28 AM	Application extens...	1,067 KB																																																																																																																																																																	
WPFXMLParser.pdb	11/20/2024 7:29 AM	VisualStudio.pdb....	162 KB																																																																																																																																																																	
zing.dll	11/20/2024 7:27 AM	Application extens...	472 KB																																																																																																																																																																	
Zxing.Net.Core.dll	11/20/2024 7:27 AM	Application extens...	70 KB																																																																																																																																																																	
Zxing.Windows.Compatibility.dll	11/20/2024 7:28 AM	Application extens...	33 KB																																																																																																																																																																	
zing-core.dll	11/20/2024 7:28 AM	Application extens...	376 KB																																																																																																																																																																	
WindowsAccessBridge-32.dll	11/21/2022 3:50 PM	Application extens...	165 KB																																																																																																																																																																	

Appium Setups (Optional)

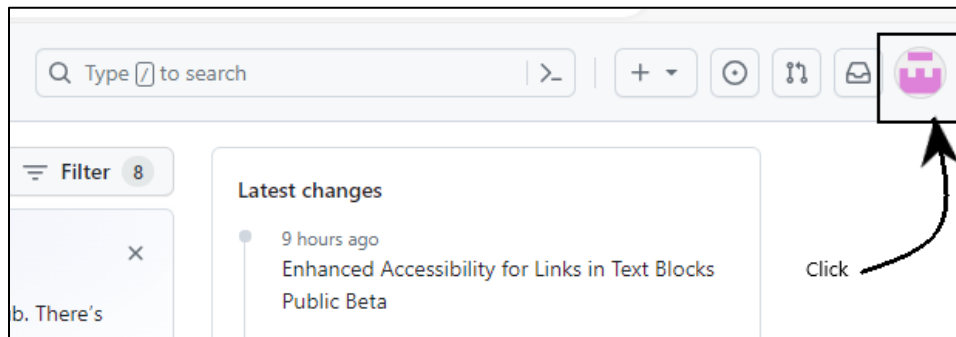
Android real and virtual device testing is explained in detail in the below document:

<https://github.com/nestedflow-Automation/NestedFlowRepo/blob/main/Documents/Mobile%20Browser%20Tests%20With%20Appium.pdf>

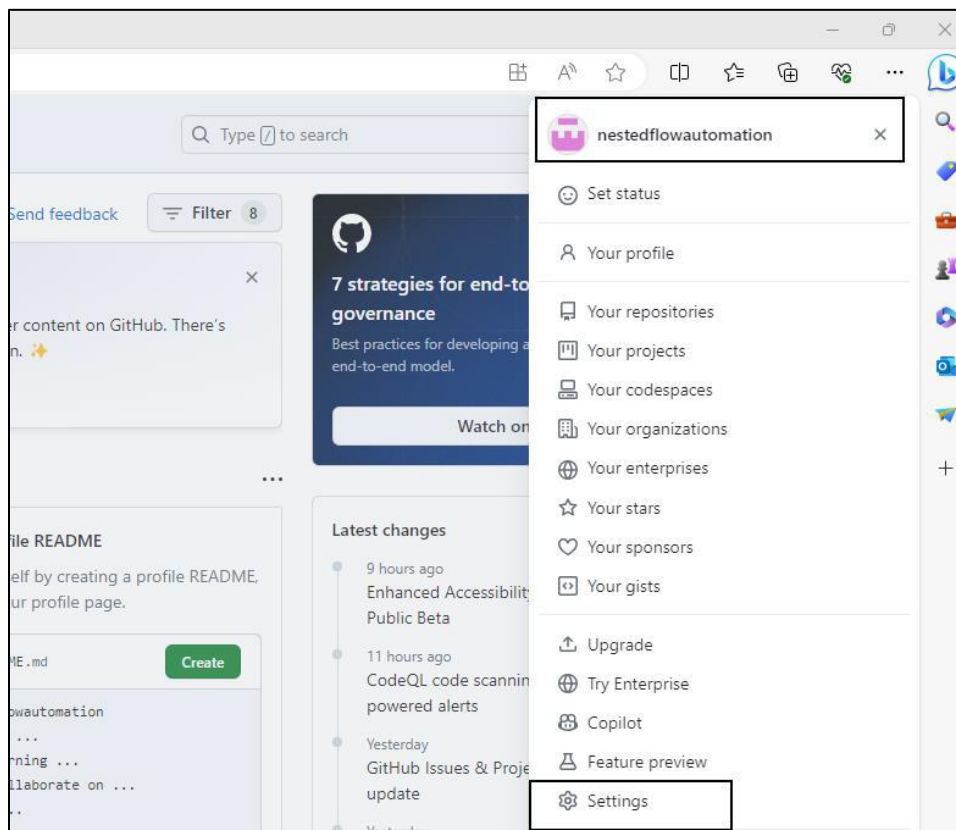
GitHub Setup (Optional)

This is an optional setup. This is needed only if the test cases are needed to be hosted in GitHub. It requires the GitHub Token to be added in configuration file.

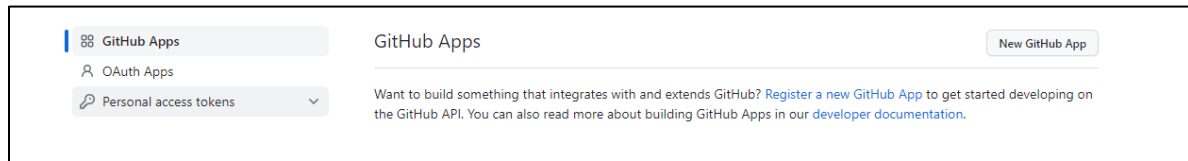
- Navigate to your Github page and login
- Navigate to settings



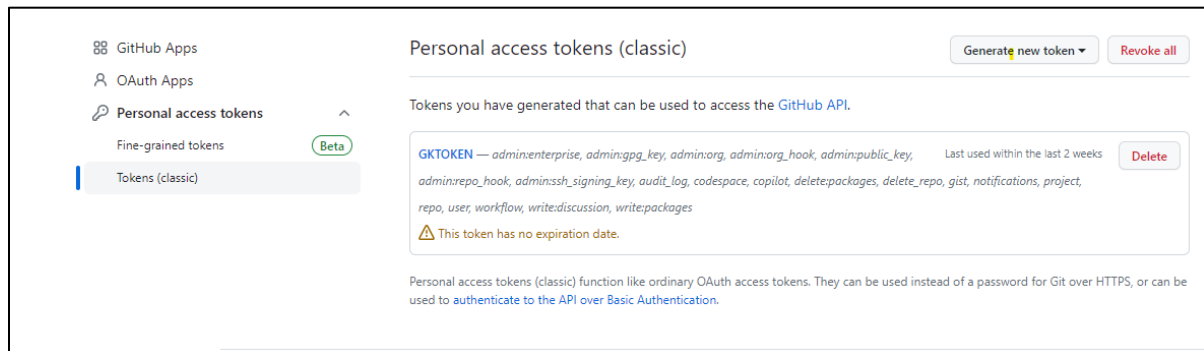
- Click on the **Developer Settings** in the bottom of the screen



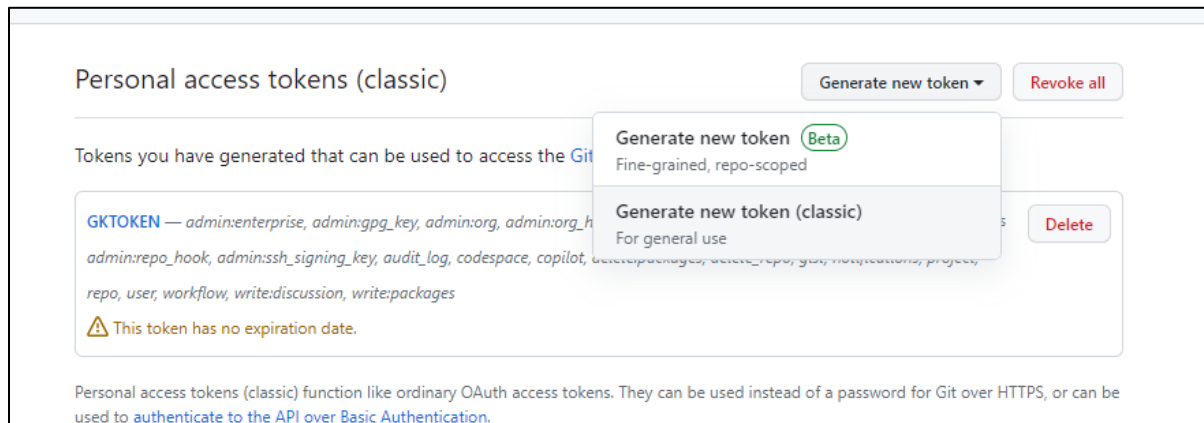
- Click on **Personal access tokens**



- Click on Tokens (Classic)



- Choose Generate new token (classic)



- If it asks for credentials enter

Confirm access

Signed in as

@nestedflowautomation

Password

[Forgot password?](#)

Confirm

Tip: You are entering **sudo mode**. After you've performed a sudo-protected action, you'll only be asked to re-authenticate again after a few hours of inactivity.

- Enter the Name for the token

- GitHub Apps
- OAuth Apps
- Personal access tokens
 - Fine-grained tokens
 - Tokens (classic)**

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

GitHubAccess

What's the token for?

Expiration *

30 days

The token will expire on Sun, Oct 29 2023

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☐ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo:deployment	Access deployment status
☐ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☐ read:org	Read org and team membership, read org projects
☐ manage_runners:org	Manage org runners and runner groups
☐ admin:public_key	Full control of user public keys

- Set Expiration to suitable value and check all possible checkboxes (both read and write rights are needed for the feature to work)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

GitHubAccess

What's this token for?

Expiration

No expiration (The token will never expire)

GitHub strongly recommends that you set an expiration date for your token to help keep your information secure. [Learn more](#)

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☒ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo:deployment	Access deployment status
☐ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events
☒ workflow	Update GitHub Action workflows
☒ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☒ delete:packages	Delete packages from GitHub Package Registry
☒ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☐ read:org	Read org and team membership, read org projects
☐ manage_runners:org	Manage org runners and runner groups
☒ admin:public_key	Full control of user public keys

- Click on Generate Token

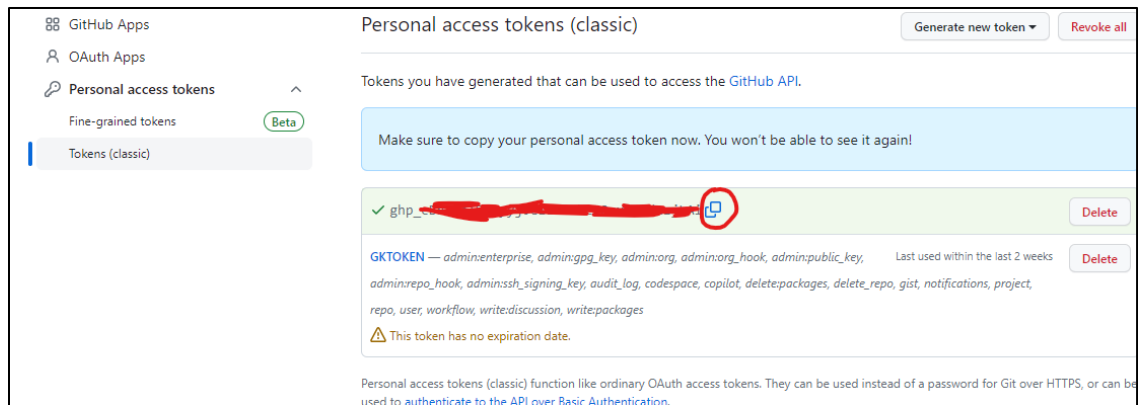
☒ write:ssh_signing_key Write p

☒ read:ssh_signing_key Read p

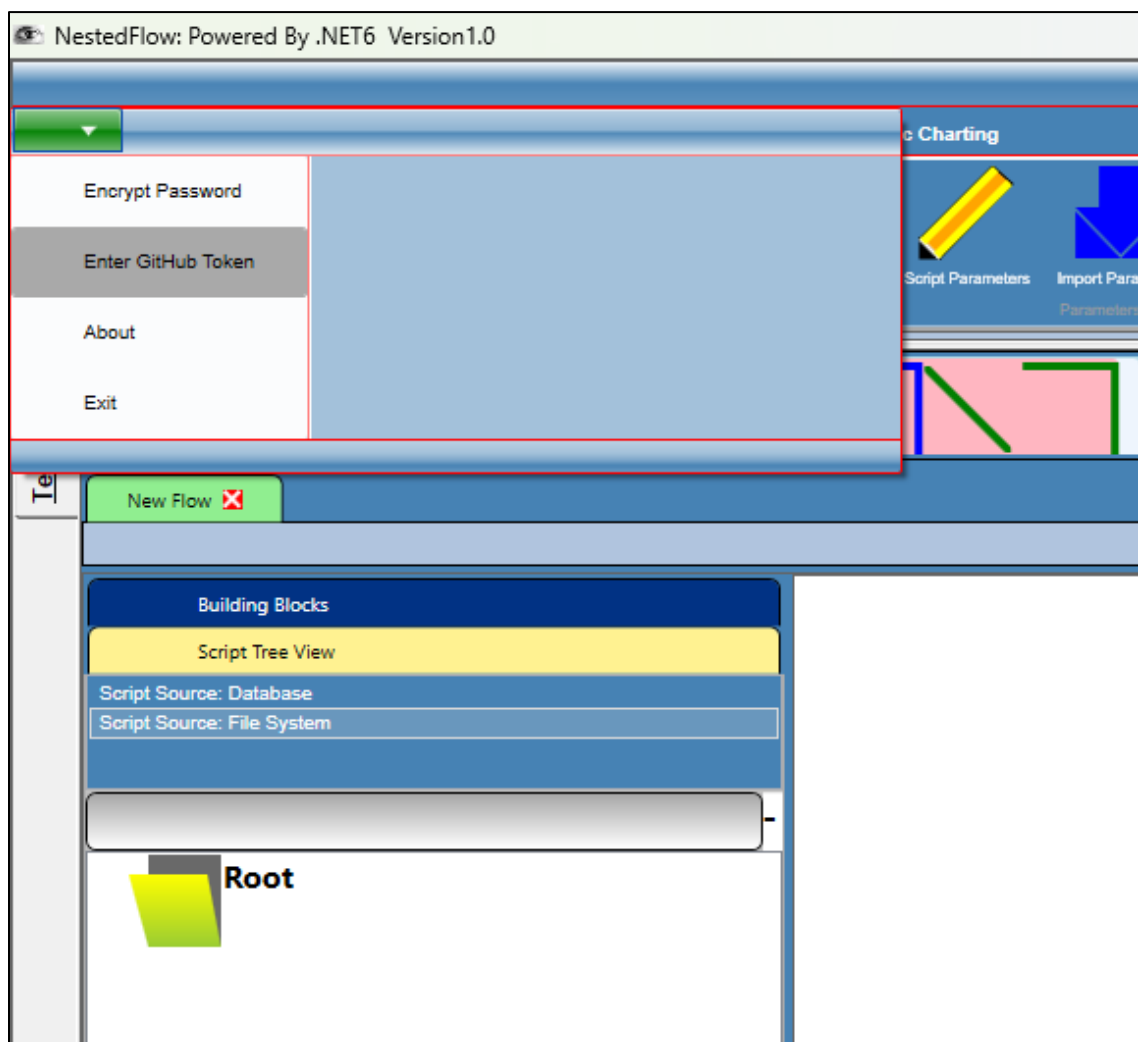
Generate token Cancel

Note: Organization Github setups will also give option to configure SSO in the token setup without which it will not work

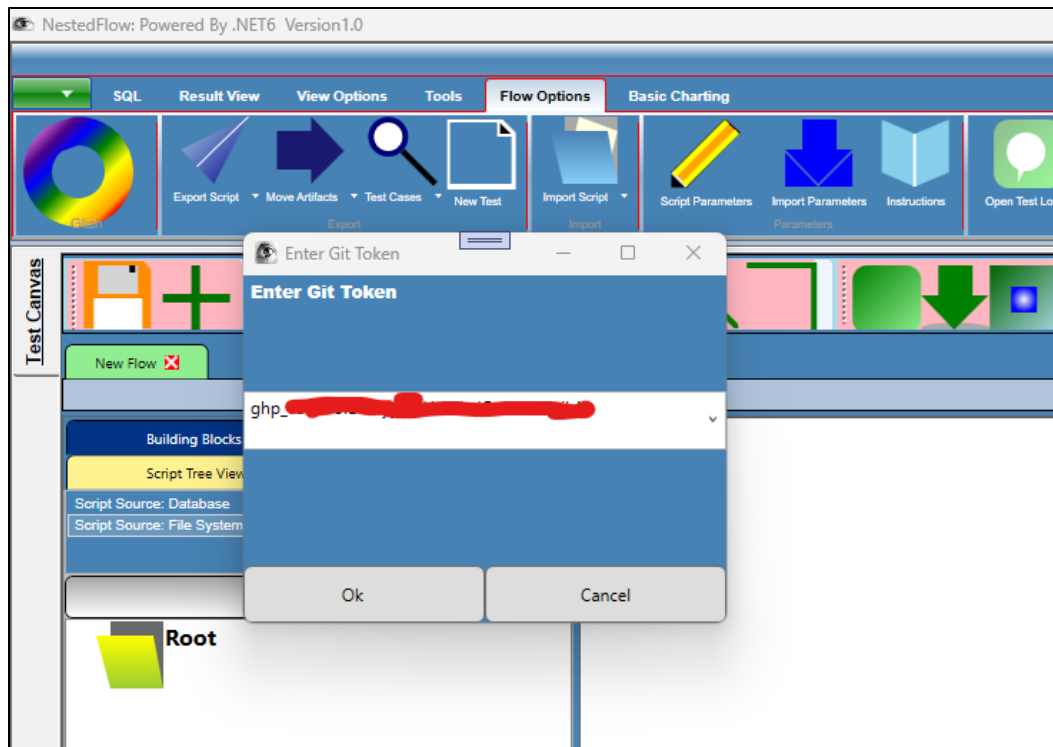
- Click on Copy token icon next to the token and it will copy to clipboard



- Open NestedFlow Automation tool. In the Ribbon menu choose Enter Github Token



- Enter Token and click Ok button



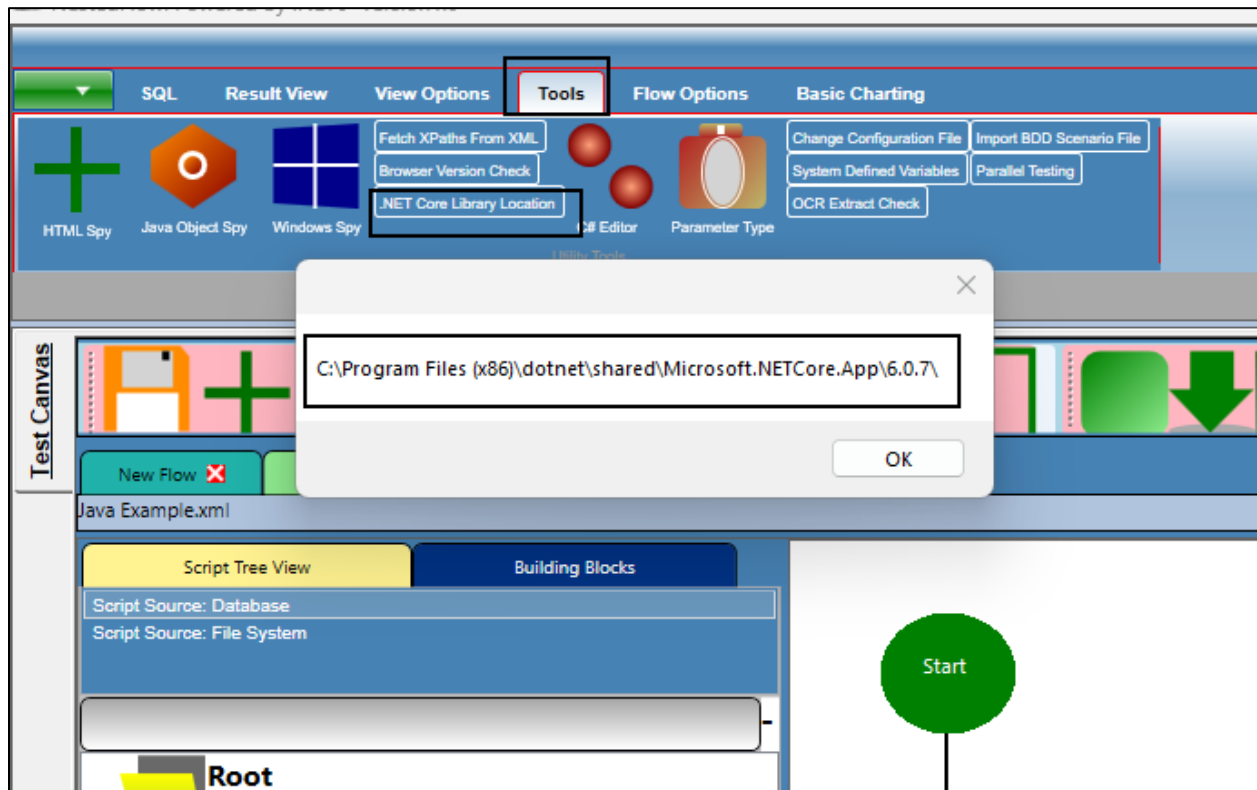
This step will update configuration file with the token.

Once this setup is done, to understand exact implementation of GitHub script hosting, please refer to the below link:

<https://github.com/nestedflow-Automation/NestedFlowRepo/blob/main/Documents/NestedFlowAutomation%20-%20Github%20Integration.pdf>

Code Considerations (optional)

NestedFlow Automation tool allows custom scripting in C# to address corner cases which are not possible to perform in scriptless way. The code follows the .NET Core standards and you can find the reference location of the libraries from **Tools** menu:



Navigate to Tools → .NET Core Library Location

Copy All the files from that location to %APPDATA%/One/CoreApp

Name	Date modified	Type	Size
.version	6/22/2022 11:31 AM	VERSION File	1 KB
api-ms-win-core-console-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-console-l1-2-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-datetime-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-debug-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-errorhandling-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-fibers-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-file-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	15 KB
api-ms-win-core-file-l1-2-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-file-l2-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-handle-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-heap-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-interlocked-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB
api-ms-win-core-libloader-l1-1-0.dll	9/11/2021 7:09 PM	Application extension...	12 KB

Note: We can as well use the standard location but If you need to use external or custom libraries you cannot use locations which only admins have access to in a machine

Add all DLLs in the NestedFlow Automation tool's binary location in Program Files (x86) should be added to %APPDATA%/One/CoreApp

All the custom DLLs you are using for the scripting should also be added to the location %APPDATA%/One/CoreApp